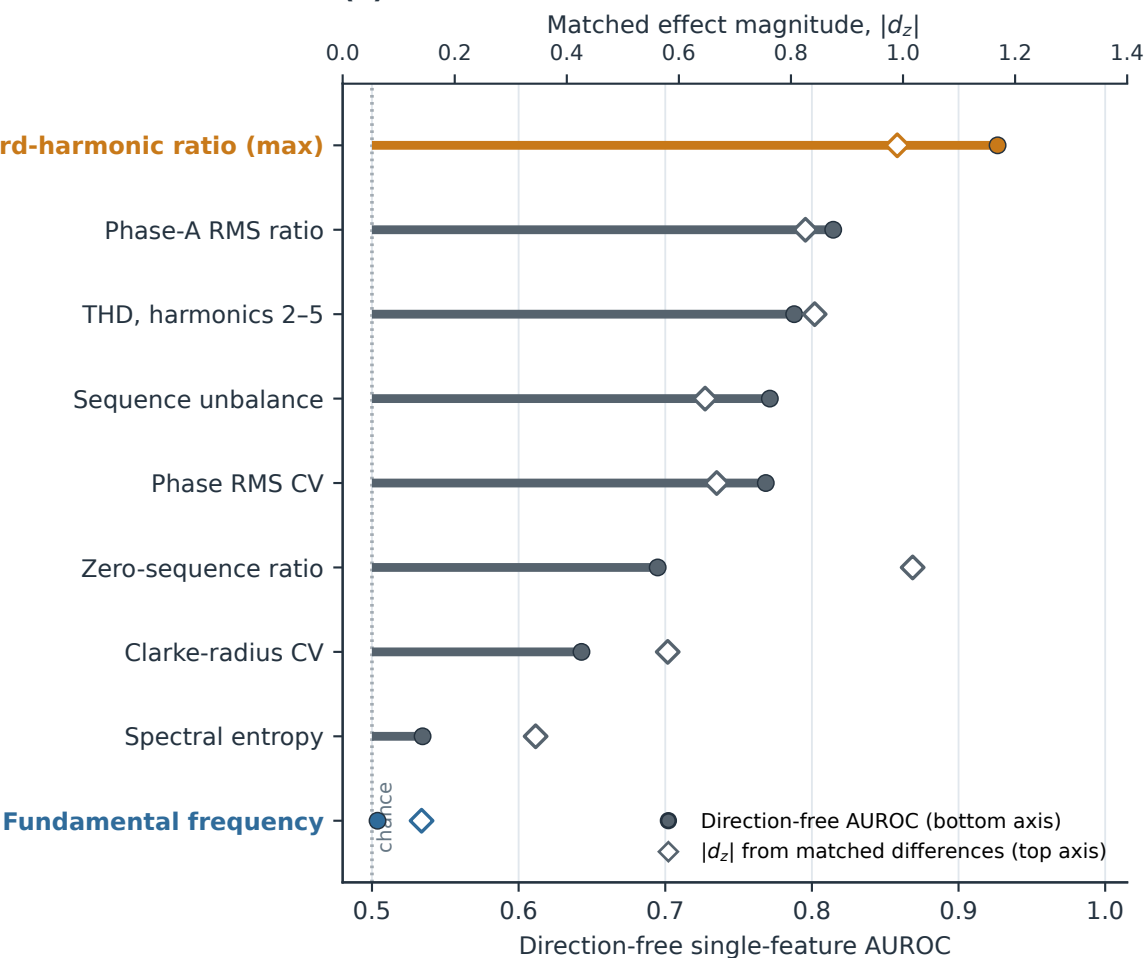


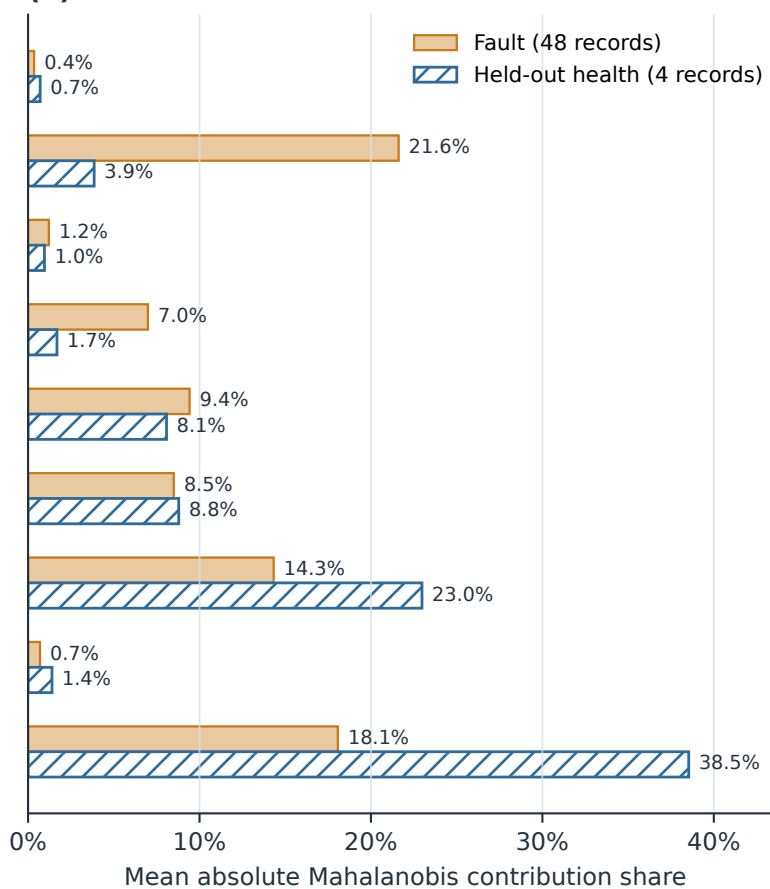
External feature discrimination and frozen-score attribution

Third-harmonic contrast separates the recorded conditions but is weakly weighted; fundamental frequency shows the inverse mismatch.

(a) Same-block discrimination



(b) Frozen-score attribution



Post-reveal exploratory diagnosis; descriptive and conditional on one physical motor. Panel (a): record $\times$ subsystem $\times$ block means at the four loads with held-out health; one health condition is reused across six fault-turn comparisons. Panel (b): winning-window attribution for all fault records versus held-out health. Cross-terms use the symmetric split $c_j = x_j(Px)$; signed contributions may be negative, while plotted shares use absolute contributions normalized to sum to 1.